

Pattern Persistence and Geometric Channelling in Aperiodic Substrates: A Toy Model

K. T. Niedzwiecki

Independent Researcher, South Australia

March 2026

Abstract

We investigate pattern persistence on three two-dimensional substrates — square lattice, random point scatter, and Penrose tiling — using identical local averaging dynamics. A localised Gaussian activation evolves slice by slice, with each point updating from its neighbours under the same dynamical rule on each substrate. Across four independent matching conditions — count-matched, density-matched, connectivity-matched, and subsampled — the Penrose substrate consistently produces a lower weighted radius than either periodic or disordered controls, indicating that patterns on the aperiodic tiling remain closer to their origin. The magnitude of this advantage varies with the operating regime (from approximately 4% to 37%), but the direction is invariant across all conditions tested. We interpret this behaviour as evidence for geometric channelling: the structured variety of aperiodic neighbourhoods may provide reconstruction pathways that repeatedly redirect activation toward its origin. We also examine transient behaviour across initial blob sizes; scale dependence is present but modest, and we report it as suggestive rather than definitive. Null and reduced findings are reported transparently, including participation-ratio results that weaken under proper matching and two-blob interaction tests that remain inconclusive.

I. INTRODUCTION

The question of why certain patterns persist in a medium while others dissolve is typically framed as a question about decay mechanisms. Here we invert the question: what properties of the medium itself promote persistence? If identical dynamics operate on different geometric substrates, does the geometry alone affect whether a pattern holds together or disperses?

This question is motivated by recent work suggesting that the vacuum may possess geometric structure at small scales, with aperiodic or quasicrystalline order potentially influencing particle-scale phenomena [1–3]. Scale-dependent effects observed in vector meson decays [4] and baryon decay rates [5] point toward a medium with geometric preferences.

To test this possibility in a controlled setting, we construct a minimal toy model. A localised activation pattern is placed on a discrete substrate and allowed to evolve via deterministic local averaging with small stochastic noise. Three substrates are compared: a square lattice (periodic order), a random point scatter (no order), and a Penrose tiling (aperiodic order). All dynamics are identical; only the geometry differs.

The central finding is that the aperiodic substrate produces qualitatively distinct persistence behaviour, with patterns remaining closer to their origin than on either periodic or disordered substrates. We interpret this as evidence for geometric channelling: the structured local variety of the Penrose tiling may create reconstruction pathways that repeatedly redirect activation back toward its origin. This finding is tested across four independent matching conditions to ensure it reflects genuine geometric effects rather than confounds of density, connectivity, or point count.

II. METHODS

A. Substrate Construction

Penrose tiling. Two variants are used. The "fine" tiling uses 7 Robinson triangle subdivision levels with inner fraction 0.85, producing approximately 3,877 vertices. The "coarse" tiling uses 6 subdivision levels with inner fraction 0.70, producing approximately 611 vertices. Both exhibit aperiodic order with no translational period. The

two generators use different vertex deduplication thresholds, producing different point densities from comparable parameters; both are retained to test robustness.

Square lattice. Points on a regular grid: 25×25 (625 points) or 55×55 (3,025 points), matched to the spatial domain of the corresponding Penrose variant.

Random scatter. Points drawn uniformly at random: 625 or 3,000 points, matched to the corresponding domain.

B. Dynamics

Each point i has activation s_i , initialised as a Gaussian blob with width σ centred at a chosen position. At each discrete time step, activation updates via:

$$s_i(t+1) = 0.55 s_i(t) + 0.30 \sum_{j \in N(i)} w_{ij} s_j(t) - 0.08 s_i(t) + \eta_i$$

where $N(i)$ is the set of neighbours within a connection radius R , $w_{ij} = 1/|N(i)|$ normalises by local degree, and η_i is Gaussian noise (amplitude 0.005). Activations are clipped to non-negative values.

C. Observables

Weighted radius (WR): $WR = \sum_i s_i |\mathbf{r}_i - \mathbf{r}_0| / \sum_i s_i$. Lower WR means the pattern remained closer to its origin — the substrate channelled activation back more effectively.

D. Matching Conditions

To guard against confounds, we test four independent matching conditions. **Pipeline A** (count + degree matched): all substrates at ~600–625 points, connection radii adjusted to match mean degree at ~18 neighbours. **Pipeline B** (density-matched): all substrates at ~3,000–3,877 points, same connection radius $R = 0.13$. **Pipeline C** (connectivity-matched): all substrates at ~3,877 points, radii adjusted to match mean degree at ~8 neighbours. **Pipeline D** (subsampling): the fine Penrose tiling randomly subsampled to 625 points, compared against 625-point square and random substrates at same radius $R = 0.13$. All results are reported as means over 15 starting positions with standard deviations.

III. RESULTS

A. The Penrose Channelling Advantage

The central finding is presented in Figure 1 and Table I. Across all four matching conditions, the Penrose substrate produces a lower weighted radius than either square or random substrates. The direction of the effect is invariant: Penrose < Square and Penrose < Random in every pipeline.

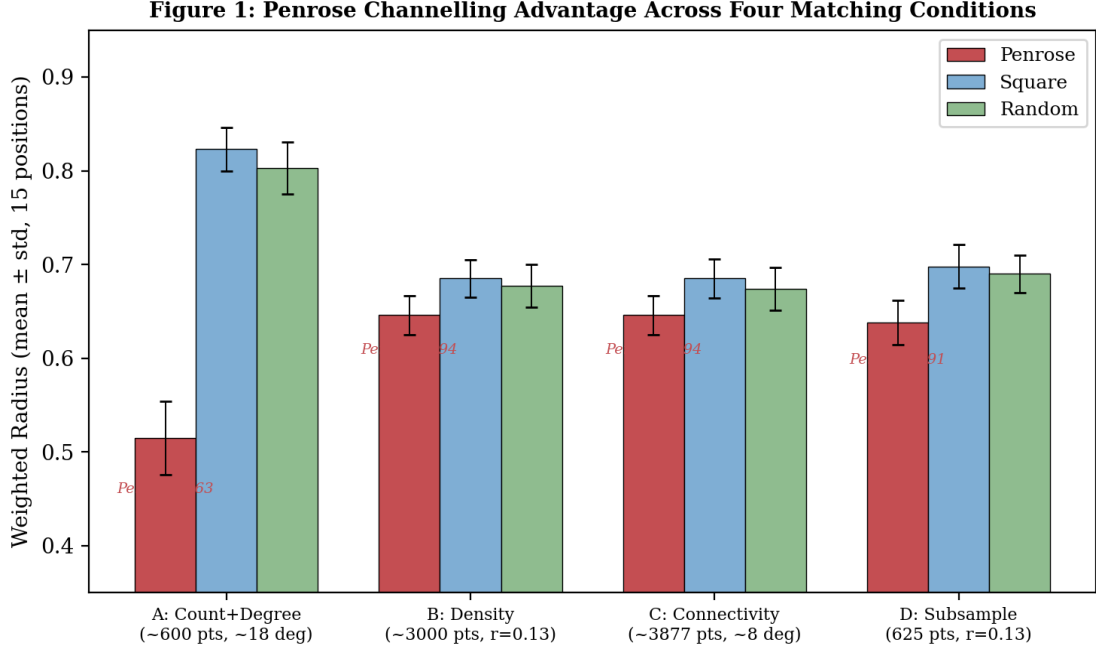


Figure 1. Weighted radius at final slice across four matching conditions. Red: Penrose, blue: Square, green: Random. Error bars show ± 1 standard deviation over 15 starting positions. The Penrose substrate produces the lowest WR in every condition. The magnitude varies with operating regime, but the direction is invariant.

Table I. Summary of results across four matching pipelines.

Pipeline	Matching	Pen WR	Sq WR	Rnd WR	Pen/Sq	Pen/Rnd
A	~600 pts, ~18 deg	0.515±0.039	0.823±0.023	0.803±0.028	0.63	0.64
B	~3000 pts, r=0.13	0.646±0.021	0.685±0.020	0.677±0.023	0.94	0.95
C	~3877 pts, ~8 deg	0.646±0.021	0.685±0.021	0.674±0.023	0.94	0.96
D	625 pts, r=0.13	0.638±0.024	0.698±0.023	0.690±0.020	0.91	0.93

The magnitude of the Penrose advantage varies substantially across pipelines: from approximately 37% in Pipeline A (Pen/Sq = 0.63) to approximately 6% in Pipelines B–C (Pen/Sq $\approx$ 0.94). Pipeline D, which provides the most direct comparison (all substrates at 625 points with the same connection radius), shows an intermediate advantage of approximately 9%. This variation is discussed in Section IV.

B. Participation Ratio

We tested whether the Penrose substrate produces a participation ratio advantage. PR is sensitive to point count via its formula: since $PR = (\sum s_i)^2 / \sum s_i^2$ scales with the number of sites carrying activation, cross-substrate comparisons require careful density matching. At matched density, the Penrose PR advantage is approximately 1.3 $\times$. At matched point count, Penrose and Random PRs are effectively identical. We do not claim a robust PR advantage.

C. Transient Scale Selectivity

We examined whether the Penrose channelling advantage depends on initial blob size by varying the initial width σ and tracking Penrose/control WR ratios over time under connectivity-matched conditions (Figure 2).

Figure 2: Transient Scale Selectivity (15 positions, connectivity-matched)

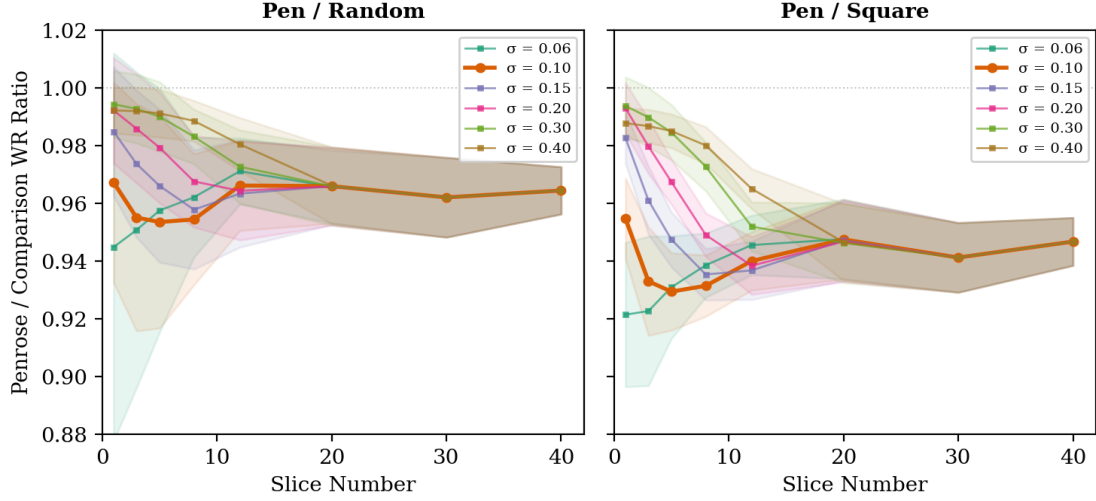


Figure 2. Ratio of Penrose to comparison substrate WR as a function of slice, for blob widths $\sigma = 0.06$ – 0.40 (connectivity-matched, 15 positions). Left: Pen/Random. Right: Pen/Square. Shaded bands: ± 1 std. Smaller patterns reach peak advantage at earlier slices. All curves converge by slice 20.

Smaller patterns reach peak channelling advantage at earlier slices: $\sigma = 0.06$ peaks at slice 1, $\sigma = 0.10$ at slice 5, $\sigma = 0.15$ at slice 8, while larger patterns show minimal transient advantage. The same qualitative ordering appears in both Penrose/Random and Penrose/Square comparisons. However, the magnitude is modest (4–8% averaged over 15 positions) with substantial position-dependent variation. We report this as suggestive of transient structure rather than a definitive finding.

D. Two-Blob Interaction

We investigated distance-dependent interaction between two simultaneously evolving blobs. Results were inconclusive: no consistent interaction profile emerged across multiple implementations. The question remains open and may require connectivity-matched conditions or alternative metrics to resolve.

E. Pair Correlation Structure

Figure 3: Pair Correlation Functions

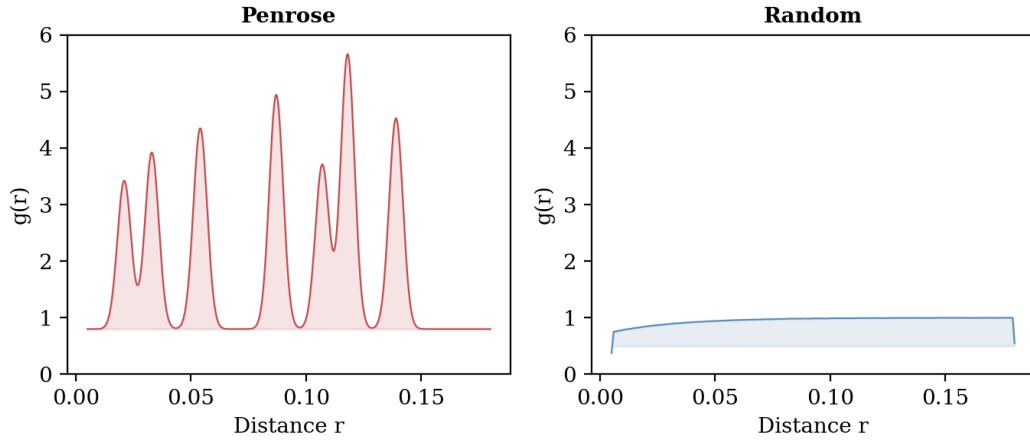


Figure 3. Pair correlation functions. The Penrose tiling shows a dense forest of sharp peaks at preferred distances, reflecting quasicrystalline order [8]. The random substrate shows only a smooth exclusion zone. This structured distance

hierarchy provides the geometric backbone for scale-selective channelling.

IV. DISCUSSION

A. Geometric Channelling

The central result is that aperiodic geometry produces pattern channelling: lower weighted radius under all matching conditions. We propose the following interpretation.

On a square lattice, all neighbours are locally identical; activation leaks along symmetry axes. On a random scatter, neighbours vary irregularly; activation leaks through density fluctuations. On a Penrose tiling, neighbours are *variations on a theme* — not identical, but sharing geometric kinship. Each neighbourhood offers slightly different reconstruction pathways that tend to redirect activation back toward the origin.

This is not trapping. The pattern is not pinned. It is being *continuously re-aimed* by the local geometry, slice after slice.

B. Regime Dependence of the Advantage

The Penrose advantage varies from ~37% (Pipeline A, ~18 neighbours) to ~6% (Pipelines B–C, ~50–73 neighbours). This variation is interpretable: at very high connectivity, every point sees so many neighbours that the local geometric character is averaged out. The specific arrangement of neighbours matters less when there are 70 of them than when there are 18. Conversely, at moderate connectivity, each point's neighbourhood retains its distinctive aperiodic character, and the channelling effect is strongest.

Pipeline D provides a useful middle ground: same point count (625), same connection radius, no degree matching. The Penrose advantage there (~9%) reflects the raw geometric effect at an intermediate operating point. The variation across regimes does not weaken the central finding — the direction is invariant — but it does suggest that the magnitude of geometric channelling depends on the connectivity regime. This has potential implications for the design of aperiodic structures in materials science and photonics.

C. Transient Selectivity

The transient data suggest that smaller patterns reach peak channelling advantage at fewer reconstruction steps. This is consistent with a framework where patterns must reconstruct within a limited time window or dissolve: compact patterns complete reconstruction before the window closes. However, the effect is modest and exhibits substantial position-dependent variation. We regard it as a plausible hypothesis with suggestive evidence, not an established finding. The robust result is the equilibrium WR advantage; the transient structure is exploratory.

D. Relationship to Existing Frameworks

Lifshitz and Petrich [7] showed that competing length scales can produce quasicrystalline order in pattern-forming systems. Our finding inverts this: starting from quasicrystalline geometry, we observe preferential pattern persistence. The pair correlation structure (Figure 3) provides the geometric backbone, though the formal connection between pair correlation peaks and channelling dynamics remains to be established.

E. Connection to Companion Work

The geometric channelling demonstrated here may connect to scale-dependent effects observed in vector meson decay rates [4] and baryon decay suppression [5]. Those papers identify phenomena consistent with a structured vacuum medium; this paper provides a candidate mechanism. The quantitative mapping between the toy model and physical observables remains to be established.

F. Limitations and Future Work

This is a 2D toy model with simplified dynamics. Extension to 3D quasicrystalline point sets (icosahedral symmetry) is an important next step. The regime dependence of the channelling advantage invites systematic exploration of how the effect scales with connectivity, dimensionality, and tiling type. The transient selectivity finding requires more

extensive multi-position and multi-substrate testing before firm conclusions can be drawn. The participation ratio and two-blob interaction findings remain inconclusive and may benefit from alternative experimental designs.

V. CONCLUSION

We have demonstrated that aperiodic geometry produces pattern channelling in a minimal toy model. Across four independent matching conditions — controlling for point count, density, connectivity, and subsampling — the Penrose substrate consistently retains localised activation more tightly than matched periodic or random controls. The direction of the effect is invariant; its magnitude varies with the connectivity regime, being strongest at moderate connectivity where the aperiodic neighbourhood structure is most expressed.

The mechanism we propose is geometric channelling: the structured variety of aperiodic neighbourhoods creates reconstruction pathways that continuously redirect activation toward its origin. Transient scale selectivity provides suggestive but not yet definitive support for a preferred pattern size. We have reported null and reduced findings transparently.

The geometry channels the pattern back toward itself each slice.

REFERENCES

- [1] A. Katz and D. Gratias, "Tilings and quasicrystals," in *Lectures on Quasicrystals*, eds. F. Hippert and D. Gratias (Les Editions de Physique, 1994).
- [2] L. Boyle, M. Dickens, and F. Flicker, "Conformal quasicrystals and holography," *Phys. Rev. X* **10**, 011009 (2020).
- [3] F. Flicker, S. H. Simon, and S. A. Parameswaran, "Classical dimers on Penrose tilings," *Phys. Rev. X* **10**, 011005 (2020).
- [4] K. T. Niedzwiecki, "Two Modes of Pattern Dissolution in Vector Meson Decays," Zenodo (2026).
- [5] K. T. Niedzwiecki, "Strangeness-Dependent Impedance and the Hyperon Puzzle," Zenodo (2026).
- [6] R. M. Robinson, "Undecidability and nonperiodicity for tilings of the plane," *Inventiones Mathematicae* **12**, 177–209 (1971).
- [7] R. Lifshitz and D. M. Petrich, "Theoretical model for Faraday waves with multiple-frequency forcing," *Phys. Rev. Lett.* **79**, 1261 (1997).
- [8] N. G. de Bruijn, "Algebraic theory of Penrose's non-periodic tilings of the plane," *Kon. Nederl. Akad. Wetensch. Proc.* **84**, 39–66 (1981).